# Daniel DeAlcala | PhD Student, Universidad Autónoma de Madrid

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## Education

**Ph.D. in Telecommunications and Computer Engineering** — **Madrid, Spain**
*Universidad Autónoma de Madrid (UAM)* — *Jan 2022 – Ongoing*

- **Awarded the FPU Fellowship**: Recipient of the most prestigious and competitive national grant for PhD training in Spain (Ministry of Science and Innovation).
- **Advisors**: Prof. Julian Fierrez and Prof. Aythami Morales.

**M.Sc. in Deep Learning for Audio and Video Signal Processing** — **Madrid, Spain**
*Universidad Autónoma de Madrid (UAM)* — *Sept 2020 – Sept 2021*

- **Thesis**: *Deep Learning for Ego-centric Video Localization and Contact Tracing*. Developed multi-user graph modeling and spatial registration algorithms for infectious disease tracking using life-logging cameras.

**B.Sc. in Telecommunications Engineering** — **Madrid, Spain**
*Universidad Autónoma de Madrid (UAM)* — *Sep 2016 – Jun 2020*

- **Academic Excellence**: **Ranked 1st in class** (Top of the 2020 graduation cohort).
- **Honors**: Awarded **8 Academic Distinctions** (Highest grade in course) and the *EPS Effort Award* for outstanding academic results in Engineering.
- **Scholarships**: Recipient of the *Excellence Scholarship*.
- **Thesis**: *Egocentric Biometric Palmprint Recognition for VR*. Designed a complete biometric authentication system and custom dataset for secure VR environments.

## R&D Projects Participation

- BBforTAI through MICINN/FEDER under Grant PID2021-127641OB-I00.
- HumanCAIC through MICINN under Grant TED2021-131787B-I00.
- M2RAI through MICIU/FEDER under Grant PID2024-160053OB-I00.
- Cátedra ENIA UAM-VERIDAS (NextGenerationEU) under Grant TSI-100927-2023-2.

## International Research Stays

**Research Intern** — **New York, USA**
*IBM Research* — *Sept – Dec 2025*

- **Researched** multi-modal end-to-end architectures for **AI-Generated Content Detection**, focusing on identifying and segmenting synthetic components (text, images, and tables) in digital documents.
- **Developed and open-sourced a generation framework** for creating large-scale synthetic datasets to support robust model training and evaluation.
- **Authored a research paper** and a **patent proposal** based on the developed methodology and framework.

## Academic Duties

Conference Reviewer.........

- Computer Vision and Pattern Recognition (CVPR), 2023,2024,2025.
- Computer Vision and Pattern Recognition Workshops (CVPR Workshops), 2024,2025.
- Association for the Advancement of Artificial Intelligence Workshops (AAAI), 2025.
- Neural Information Processing Systems (NeurIPS), 2025.
- International Conference on Computer Vision (ICCV), 2025.

Teaching.........

- Multimedia Signal Processing, B.Sc. in Telecommunications Engineering (2023 - Present)

## Publications


[1] R. Daza, D. DeAlcala, A. Morales, R. Tolosana, R. Cobos, and J. Fierrez, "ALEBk: Feasibility Study of Attention Level Estimation via Blink Detection applied to e-Learning," *AAAI Workshop on Artificial Intelligence for Education (AI4EDU)*, 2022.

[2] I. Serna, D. DeAlcala, A. Morales, J. Fierrez, and J. Ortega-Garcia, "IFBiD: Inference-Free Bias Detection," *AAAI Workshop on Artificial Intelligence Safety (SafeAI)*, 2022.

[3] D. DeAlcala et al., "BeCAPTCHA-Type: Biometric Keystroke Data Generation for Improved Bot Detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR)*, 2023, pp. 1051–1060.

[4] D. DeAlcala, I. Serna, A. Morales, J. Fierrez, and J. Ortega-Garcia, "Measuring Bias in AI Models: An Statistical Approach Introducing N-Sigma," in *2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC)*, IEEE, 2023, pp. 1167–1172.

[5] D. DeAlcala, G. Mancera, A. Morales, J. Fierrez, R. Tolosana, and J. Ortega-Garcia, "A Comprehensive Analysis of Factors Impacting Membership Inference," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR)*, 2024, pp. 3585–3593.

[6] D. DeAlcala, A. Morales, J. Fierrez, G. Mancera, R. Tolosana, and J. Ortega-Garcia, "Is My Data in Your AI? Membership Inference Test (MINT) applied to Face Biometrics," *IEEE Access*, 2025.

[7] D. DeAlcala, A. Morales, J. Fierrez, G. Mancera, R. Tolosana, and R. Vera-Rodriguez, "MINT-Demo: Membership Inference Test Demonstrator," *AAAI Workshop on Artificial Intelligence Governance (AIGOV)*, 2025.

[8] G. Mancera, D. DeAlcala, J. Fierrez, R. Tolosana, and A. Morales, "Is My Text in Your AI Model? Gradient-based Membership Inference Test applied to LLMs," *arXiv preprint arXiv:2503.07384*, 2025.

[9] D. DeAlcala, A. Morales, J. Fierrez, G. Mancera, and R. Tolosana, "gMINT: Gradient-based Membership Inference Test applied to Image Models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR)*, 2025, pp. 2781–2790.

[10] D. DeAlcala, A. Morales, J. Fierrez, G. Mancera, R. Tolosana, and J. Ortega-Garcia, "Active Membership Inference Test (aMINT): Enhancing Model Auditability with Multi-Task Learning," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 647–656.

[11] D. DeAlcala, A. Morales, J. Fierrez, and R. Tolosana, "AttZoom: Attention Zoom for Better Visual Features," in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCV)*, 2025, pp. 4834–4843.

[12] P. Korshunov et al., "DeepID Challenge of Detecting Synthetic Manipulations in ID Documents," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 510–519.

[13] D. DeAlcala, K. Ching-Yun, A. Morales, J. Fierrez, R. Tolosana, and P.-Y. Chen, "AIDoc: A Framework for Generating and Identifying AI-Edited Documents," in *Under Review in International Conference on Machine Learning (ICML)*, 2026.


## Participation to Conferences

- Computer Vision and Pattern Recognition Conference (CVPR), 2023. Vancouver, Canada.
- Computers, Software, and Applications Conference (COMPSAC), 2023. Torino, Italy.
- Computer Vision and Pattern Recognition Conference (CVPR), 2024. Seattle, WA, USA.
- The Association for the Advancement of Artificial Intelligence (AAAI), 2025. Philadelphia, Pennsylvania, USA.
- Computer Vision and Pattern Recognition Conference (CVPR), 2025. Nashville, TN, USA.
- International Conference on Computer Vision (ICCV), 2025. Honolulu, Hawaii.